IBVAP ko Mentor ko Kaise Samjhayein

SIH 2026 ki poori PPT, Hinglish mein, slide by slide

Problem Statement	PS26187 - AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure
Theme	Blockchain & Cybersecurity
Category	Software
Team	Vision Vigilants

> Is guide mein har slide ka matlab hai aur bolne ka script bhi. Script ko ratna nahi hai, apne shabdon mein bolna hai. Aakhir mein do page sirf aapke liye hain: blockchain wale sawaal ka jawaab, aur ek honest list ki kya bana hai aur kya nahi.

SHURUAAT

Sabse Pehle: 30 Second Ka Opening

Mentor ke saamne baithte hi poori technical baat mat shuru karna. Pehle 30 second mein yeh teen cheezein bolni hain: dikkat kya hai, humne kya banaya hai, aur usse farak kya padta hai. Neeche wala paragraph ratna nahi hai, bas iska matlab yaad rakhna hai.

OPENING SCRIPT

Sir, hamara problem statement hai PS26187. Border par pehle se hazaaron CCTV camera lage hue hain aur woh chaubees ghante recording karte rehte hain. Dikkat camera ki nahi hai. Dikkat yeh hai ki us footage ko dekhne wala ek insaan hai. Koi bhi aadmi bees pachees minute tak screen par lagataar dhyaan nahi de sakta, yeh insaan ki kudrati seema hai. Isliye footage record to ho rahi hai, lekin uska faayda tab hota hai jab ghatna ho chuki hoti hai aur hum peeche jaake dekhte hain.

Hamara solution IBVAP hai, matlab Intelligent Border Video Analytics Platform. Hum naya camera nahi lagate. Hum usi purane CCTV ki video stream ko AI se padhte hain, aur jab kuch galat lagta hai tabhi operator ko alert bhejte hain. Matlab jawaan ko ab chaubees ghante camera nahi dekhna, sirf alert dekhna hai.

Ek line mein: recording se intelligence banaya hai, aur hardware ka kharcha zero rakha hai.

SLIDE 1 OF 6

TITLE PAGE

Is slide ka kaam

Yeh slide sirf pehchaan ke liye hai. Iska matlab hai ki judge ko pata chal jaye ki aap kaunsa problem statement utha rahe ho. Ispar 20 second se zyada mat lagana.

DHYAAN RAKHNA

Agar Team ID abhi blank hai to submission se pehle bhar lena. Judge ke saamne khaali field achha nahi lagta.

Slide par kya likha hai

- Problem Statement ID: PS26187
- Title: AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure
- Theme: Blockchain & Cybersecurity
- Category: Software
- Team Name: VisionVigilants

AAPKO KYA BOLNA HAI

Sir, hum team Vision Vigilants hain aur humne problem statement PS26187 uthaya hai. Iska title hai AI based intelligent video analytics platform for border surveillance using existing CCTV infrastructure.

Is title mein sabse important shabd hai existing. Problem statement kehta hai ki naya system banao lekin jo camera pehle se lage hue hain unhi par banao. Yeh hamari poori design ki buniyaad hai, aur baaki saare slides isi ek shart ko follow karte hain.

Category software hai, matlab hum koi naya hardware nahi bech rahe. Theme blockchain aur cybersecurity hai, aur uspar main slide teen ke baad detail mein aata hoon.

SLIDE 2 OF 6

IBVAP - IDEA AUR USP

Is slide ka kaam

Yeh slide aapka main pitch hai. Isme teen hisse hain: Pillars of Innovation, Unique Selling Proposition, aur Stakeholders. Judge yahin decide karta hai ki aage sunna hai ya nahi. Ispar 2 minute do.

DHYAAN RAKHNA

Agar judge poochhe ki USP kya hai to ek line yaad rakho: hum naya camera nahi maangte, isliye hamari deployment cost sirf software ki hai.

Slide par kya likha hai

- Upar ek box: existing CCTV bahut data banata hai, par manual monitoring se time par khatra pakad mein nahi aata. IBVAP usi feed ko intelligent bana deta hai. Passive monitoring se proactive threat awareness.
- Pillars of Innovation: alag alag border scenario samajhna, aam harkat aur shaq wali harkat mein farak karna, aur ek jagah par poora threat picture banana.
- USP ke chaar card: Legacy CCTV Integration, Hardware-Agnostic AI, Real-Time Analytics, Scalable Multi-Camera.
- Stakeholders: SSB aur border forces, BOP personnel, MHA, aur IT teams.

AAPKO KYA BOLNA HAI

Sir, upar wala box hamari poori soch hai. Aaj border par jo CCTV lage hain woh data to bahut banate hain, lekin woh data kisi kaam ka tab hi hai jab koi use time par dekh le. Practically woh hota nahi hai. IBVAP us footage ko machine se padhwata hai, aur insaan ko sirf tab bulata hai jab zaroorat ho.

Pillars of Innovation mein teen baatein hain. Pehli, border ek jaisa nahi hota. Kahin fencing hai, kahin khuli zameen hai, kahin road aur checkpoint hai. Hamara system teenon mein kaam karta hai. Doosri, har harkat khatra nahi hoti. Gaon ka aadmi roz us raste se jaata hai, woh normal hai. Hum routine aur suspicious mein farak karte hain. Teesri, alag alag camera ke alag alag alert ko jodkar ek hi threat picture banate hain, taaki officer ko poori kahani dikhe na ki tukde.

USP ke chaar card mein sabse zaroori do hain. Legacy CCTV Integration matlab jo camera lage hain wohi chalenge, kuch badalna nahi. Aur Hardware-Agnostic AI matlab hamein koi special AI camera nahi chahiye. Hum RTSP stream lete hain, jo har IP camera aur har NVR deta hai. Isliye brand se koi farak nahi padta.

Stakeholder wale hisse mein main yeh point karta hoon ki hamare do alag alag user hain aur unki zaroorat alag hai. BOP par baitha jawaan turant alert chahta hai. Headquarters baitha officer trend aur history chahta hai. Hamare system mein dono ke liye alag view hai, aur dono ke liye alag login role hai.

MUSHKIL SHABDON KA MATLAB

RTSP	Real Time Streaming Protocol. Ek standard tareeka jisse camera apni live video network par bhejta hai. Har IP camera ismein video deta hai.
ONVIF	Camera companies ka common standard, taaki alag alag brand ke camera ek hi software se chal sakein.
NVR / DVR	Network ya Digital Video Recorder. Woh box jisme camera ki recording jama hoti hai. Hum yahan se bhi feed le sakte hain.
BOP	Border Outpost. Border par jawaano ki chhoti post jahan se nigraani hoti hai.

SLIDE 3 OF 6

TECHNICAL APPROACH

Is slide ka kaam

Yeh sabse technical slide hai aur yahin sabse zyada sawaal aate hain. Diagram ko baayen se daayen padho: video andar aati hai, beech mein AI use samajhta hai, daayen taraf operator ko result milta hai. Ispar 2 se 3 minute do.

DHYAAN RAKHNA

Yeh slide par likha hai 'A Multi-Layered AI & Blockchain Framework'. Blockchain abhi code mein nahi hai. Iska poora jawaab is guide ke aage wale section mein hai, woh zaroor padh lena.

Slide par kya likha hai

- Baayen taraf Video Input Sources: IP camera, RTSP ya ONVIF stream, purana NVR ya DVR, aur kayi BOP ke alag alag source.
- Beech mein AI Video Analytics Pipeline: frames, pre-processing (OpenCV), object detection (YOLOv8), tracking (ByteTrack), pose estimation (YOLOv8-pose), ANPR, aur anomaly detection (Isolation Forest).
- Uske aage Behaviour aur Threat Analysis, jo threat_rules.yaml se chalta hai, aur Event Correlation and Scoring.
- Daayen taraf do raste: normal activity sirf log hoti hai, suspicious activity se incident banta hai, alert jaata hai aur evidence store hota hai.
- Neeche ki patti: AI models, training aur validation, deployment, API, aur modular design.

AAPKO KYA BOLNA HAI

Sir, is diagram ko main baayen se daayen padhta hoon.

Sabse baayen video aati hai. Hum koi khaas camera nahi maangte. Jo IP camera hai, jo RTSP stream deta hai, ya jo purana DVR pehle se laga hai, teenon se feed le sakte hain.

Beech wala gulaabi hissa hamara dimaag hai. Yahan chaar kaam line se hote hain. Pehla, YOLOv8 se object detection: frame mein aadmi hai, gaadi hai, bag hai, yeh pata chalta hai. Doosra, ByteTrack se tracking: pata chalta hai ki yeh wahi aadmi hai jo pichle frame mein tha, aur usko ek ID mil jaati hai. Teesra, pose estimation: hum aadmi ke sharir par sattrah point nikaalte hain, kandha, kohni, ghutna, aur usse pata chalta hai ki woh khada hai, jhuka hua hai, ya deewar chadh raha hai. Chautha, gaadi ke liye ANPR, matlab number plate padhna.

Iske baad threat analysis hota hai. Yeh hamara rule engine hai aur yeh ek YAML file se chalta hai jiska naam threat_rules.yaml hai. Ismein solah alag alag factor hain, jaise raat ka time, fence ke paas rukna, ulti disha mein chalna, group banana. Har factor ka apna weight hai aur sabko jodkar zero se sau ke beech ek score banta hai. Us score ke hisaab se khatra LOW, MEDIUM, HIGH ya CRITICAL hota hai.

Yahan main ek design choice zaroor batata hoon. Humne rule engine ko YAML file mein rakha hai, code ke andar nahi. Iska matlab yeh hai ki agar kisi border sector mein raat ki harkat aam baat hai, to wahan ka officer bina programmer ke, bina naya software banaye, us factor ka weight badal sakta hai. Yeh field mein bahut kaam aata hai.

> Daayen taraf do raste hain. Agar activity normal hai to bas log ho jaati hai, koi alert nahi. Agar suspicious hai to incident banta hai, operator ko alert jaata hai, aur us waqt ki image aur clip evidence ke taur par save ho jaati hai. Yeh evidence isliye zaroori hai kyunki baad mein enquiry mein proof chahiye hota hai.

MUSHKIL SHABDON KA MATLAB

YOLOv8	Ek object detection model. Ek hi baar image dekhkar bata deta hai ki usme kya kya hai aur kahan hai. Isliye yeh live video ke liye tez hai.
ByteTrack	Tracking algorithm. Detection ko jodkar batata hai ki alag alag frame mein dikhne wala aadmi ek hi aadmi hai.
Pose Estimation	Aadmi ke sharir ke jodon ke point nikaalna, taaki uski harkat samajh aaye, sirf uska box nahi.
ANPR	Automatic Number Plate Recognition. Gaadi ka number plate padhna.
Isolation Forest	Ek machine learning tareeka jo bina sikhaye anokhi cheez pakadta hai. Isko yeh nahi batana padta ki galat kya hai, yeh khud dekh leta hai ki yeh baaki sab se alag hai.
OpenCV	Image aur video par kaam karne ki standard library.
FastAPI	Python ka web framework. Hamara backend isi par bana hai aur React dashboard isi se data leta hai.

SLIDE 4 OF 6

FEASIBILITY AND VIABILITY

Is slide ka kaam

Yeh slide judge ko batata hai ki aapka idea sirf kaagaz par achha nahi hai, balki asli mein ban sakta hai aur chal sakta hai. Yahan aapko honest dikhna hai, kyunki aapne khud challenges likhe hain. Ispar 2 minute do.

DHYAAN RAKHNA

Challenges wale hisse ko kamzori mat samajhna. Judge ko woh team achhi lagti hai jo apni dikkat khud jaanti hai. Jo team kehti hai ki koi problem nahi hai, uspar shaq hota hai.

Slide par kya likha hai

- Feasibility ke chaar point: existing CCTV compatible, proven AI technologies, modular architecture, aur edge ya server dono par deployment.
- Viability ke chaar point: cost effective deployment, model time ke saath behtar hota rahega, operator feedback se false positive kam honge, aur naye module bina poora system badle jod sakte hain.
- Challenges aur unki strategy: low light ke liye enhancement aur RGB-thermal fusion, false positive ke liye temporal aur tracking based validation, real time scalability ke liye model optimization aur adaptive frame sampling.
- Daayen taraf scale: 2450 km SSB ka Indo-Nepal aur Indo-Bhutan border, Rs 10,985 Cr SSB allocation, Rs 13,020 Cr BIM scheme, aur 6,324 log pakde gaye.

AAPKO KYA BOLNA HAI

Sir, feasibility wale hisse mein sabse bada point yeh hai ki humne koi naya research model nahi banaya. Humne woh models use kiye hain jo already proven hain aur jinke paper published hain. Naya kaam humne unko border ke context mein jodne mein kiya hai. Isliye risk kam hai aur banne ka time kam hai.

Modular architecture ka matlab simple hai. Har analytics module alag hai. ANPR alag file hai, intrusion alag, pose alag. Agar kal ko koi ek module band bhi ho jaye to baaki system chalta rahega, aur naya module jodne ke liye poora system dobara nahi likhna padega.

Viability mein main yeh bolta hoon ki system time ke saath behtar hota hai. Jab operator kisi alert ko dekhkar batata hai ki yeh galat alarm tha, to woh feedback log ho jaata hai. Us data se hum thresholds tune kar sakte hain.

Ab challenges. Main yeh khud bolta hoon, judge ke poochhne ka intezaar nahi karta, kyunki isse bharosa banta hai. Teen asli dikkat hain.

Pehli, raat ka andhera. Border ki nigraani ka asli kaam raat mein hi hota hai aur wahin camera sabse kamzor hota hai. Humne abhi low light detection ka module banaya hai jo pehchaan leta hai ki frame kitna andhera hai aur uske hisaab se image ko enhance karta hai. RGB aur thermal camera ko milane wali baat hamari strategy hai, woh abhi bani nahi hai, kyunki uske liye thermal camera chahiye jo hamare paas test ke liye nahi hai.

Doosri, false positive. Yeh sabse asli khatra hai. Agar system har jhaadi hilne par alert de dega to teen din mein jawaan alert dekhna band kar dega, aur phir poora system bekaar ho jayega. Isliye humne alert dene ka tareeka aisa rakha hai ki ek frame par kabhi faisla nahi hota. Kisi bhi track ko pehle kam se kam teen baar dikhna padta hai, aur boundary ke paas wali harkat pehle arm hoti hai, phir agle frames mein confirm hone par hi alert banti hai.

Teesri, speed. Agar hum har frame par poora AI chalayein to ek machine par chaar camera bhi nahi chalenge. Isliye hum har doosra frame process karte hain, aur pose estimation sirf un logon par chalate hain jo kaafi bade dikh rahe hain, door ke chhote figure par nahi. Isse quality mein farak nahi padta lekin load aadha ho jaata hai.

Daayen taraf ke numbers ke liye main sirf itna bolta hoon ki yeh problem chhoti nahi hai. Akela SSB 2450 km border dekhta hai, aur sarkar border technology par hazaaron crore laga rahi hai. Hamara point yeh hai ki us paisa ka ek hissa software se bhi nikala ja sakta hai, kyunki camera to already lage hue hain.

MUSHKIL SHABDON KA MATLAB

False Positive	Galat alarm. Jab system kehta hai ki khatra hai lekin asal mein kuch nahi tha.
Edge Deployment	AI ko wahin chalana jahan camera hai, BOP par hi. Iska faayda yeh hai ki internet slow ho to bhi kaam chalta rahe.
Frame Sampling	Har frame ke bajaye beech beech ke frame process karna, taaki machine par load kam rahe.
RGB-Thermal Fusion	Normal camera aur garmi dekhne wale thermal camera ki image ko milana, taaki raat mein bhi aadmi dikhe.

SLIDE 5 OF 6

IMPACT AND BENEFITS

Is slide ka kaam

Yeh slide batata hai ki iska faayda kise hoga. Isko technical mat banao. Yahan aapko insaan ki baat karni hai, code ki nahi. Ispar 1 se 2 minute do.

DHYAAN RAKHNA

Yahan ek asli example dena bahut asar karta hai. Jaise: das ghante ki recording mein se kisi ek aadmi ko dhoondhna aaj manual kaam hai, hamare system mein woh search ban jaata hai.

Slide par kya likha hai

- Benefits for BOP Personnel: dimaag par bojh kam, alert aur playback se verification tez, zaroori event pehle dikhte hain.
- Benefits for Border Forces: kayi camera ki automated nigraani, intrusion aur suspicious activity ka real time alert, aur ek jagah par timestamp ke saath poora record.
- Benefits for Authorities: kayi BOP ka ek saath view, history se planning, aur bina camera badle scale karna.
- Chaar impact box: operational, economic, social and security, aur sustainability.

AAPKO KYA BOLNA HAI

Sir, is slide par main sabse pehle BOP ke jawaan ki baat karta hoon, kyunki asli user wohi hai.

Aaj uska kaam hai screen dekhte rehna. Yeh kaam thakane wala hai aur ismein galti insaan ki nahi, dhyaan ki seema ki hoti hai. Hamare system ke baad uska kaam badal jaata hai. Ab woh screen nahi dekhta, woh alert verify karta hai. Alert ke saath us waqt ki image aur clip already attached hoti hai, to use ghanton ki recording mein peeche jaakar dhoondhna nahi padta. Jo kaam pehle das minute ka tha woh das second ka ho jaata hai.

Force level par faayda yeh hai ki nigraani ab ek aadmi ke dhyaan par nahi tiki. Chaar camera ho ya chalis, system sab par ek jaisa dhyaan deta hai, aur raat ke teen baje bhi utna hi deta hai jitna din mein.

Authorities ke liye sabse bada faayda history hai. Aaj har BOP ka record alag hota hai. Hamare system mein sab ek database mein aata hai, to officer dekh sakta hai ki pichle teen mahine mein kis sector mein kis time par sabse zyada harkat hui. Isse patrolling ki planning data ke hisaab se hoti hai, andaaze se nahi.

Economic impact mein ek line important hai. Kharcha isliye kam hai kyunki camera pehle se laga hua hai. Hum sirf software aur ek server ki baat kar rahe hain. Aur scale karna bhi asaan hai, ek baar mein poora border nahi karna padta, do BOP se shuru karke badhaya ja sakta hai.

SLIDE 6 OF 6

RESEARCH AND REFERENCES

Is slide ka kaam

Yeh slide dikhata hai ki aapne padha hai, sirf soch kar nahi likha. Ispar 30 second se zyada mat lagana, lekin har reference ka matlab pata hona chahiye. Agar judge kisi ek par ungli rakh de aur aap na bata pao to bura lagta hai.

DHYAAN RAKHNA

Har reference kholkar ek baar dekh lena, khaas karke MHA aur PIB wale link. Agar judge poochhe ki CIBMS kya hai aur aap na bata pao to reference ulta pad jaata hai.

Slide par kya likha hai

- Reference 1 se 3: sarkari source. MHA ka Border Management division aur CIBMS, PIB ka smart fencing release, aur Sashastra Seema Bal Act 2007.
- Reference 4: YOLO ka original paper, Redmon et al, CVPR 2016.
- Reference 5: ByteTrack ka paper, Zhang et al, 2022.
- Reference 6 aur 8: low light aur raat ke dataset, LLVIP aur NightOwls.
- Reference 7: MOT17, tracking ka standard benchmark dataset.

AAPKO KYA BOLNA HAI

Sir, hamare references do hisson mein hain.

Pehle teen sarkari source hain. MHA ka border management division aur CIBMS isliye diya hai kyunki CIBMS pehle se ek sarkari programme hai jismein border par sensor aur camera lagaye ja rahe hain. Hamara system uske khilaf nahi, uske upar baithta hai. SSB Act isliye diya hai taaki pata chale ki zimmedari kiski hai aur user kaun hai.

Baaki references technical hain. YOLO aur ByteTrack ke original paper isliye diye hain kyunki hamara detection aur tracking unhi par bana hai. LLVIP aur NightOwls raat ke dataset hain, aur MOT17 tracking ka standard benchmark hai. Yeh dikhata hai ki raat wali dikkat hum jaante hain aur duniya mein uspar kaam ho raha hai.

MUSHKIL SHABDON KA MATLAB

CIBMS	Comprehensive Integrated Border Management System. Sarkar ka programme jismein border par sensor, camera aur command centre lagaye ja rahe hain.
SSB	Sashastra Seema Bal. Yeh force Indo-Nepal aur Indo-Bhutan border sambhalti hai.
MOT17	Multi Object Tracking ka standard dataset, jispar tracking algorithms ki performance naapi jaati hai.

ZAROORI

Blockchain Wala Sawaal: Iska Jawaab Tayyar Rakho

YEH PEHLE PADHO

Slide 3 ke neeche likha hai: A Multi-Layered AI & Blockchain Framework. Theme bhi Blockchain & Cybersecurity hai. Lekin project ke code mein blockchain kahin nahi hai. Maine poora codebase check kiya, blockchain ya hash chain ka ek bhi module nahi hai. Agar mentor ne yeh pakad liya aur aapne ghuma phira kar jawaab diya to bharosa toot jaayega. Isliye seedha aur tayyar jawaab yeh hai.

MENTOR POOCHHE TO YEH BOLNA

Sir, theme blockchain aur cybersecurity hai. Main saaf bata deta hoon ki hamara kaam abhi cybersecurity wale hisse par hua hai, blockchain wale par nahi.

Cybersecurity mein jo bana hua hai woh yeh hai. Password kabhi plain text mein store nahi hota, woh scrypt se hash hota hai. Scrypt isliye chuna kyunki woh jaan bujhke dheema hai, to koi chori kiya hua database leke crack nahi kar sakta. Login ka session token bhi seedha store nahi hota, uska sirf fingerprint store hota hai. Do role hain, Super Admin aur Normal User, aur permission frontend mein nahi balki backend mein check hoti hai, taaki koi browser ka code badalkar bypass na kar sake. Aur har zaroori kaam ka audit log banta hai: kisne login kiya, kisne login try karke fail kiya, kisne camera band ki, aur kisne evidence export kiya.

Blockchain abhi implement nahi kiya hai, aur main galat nahi bolna chahta. Lekin uski jagah hamare system mein ekdum saaf hai, aur woh yeh hai. Jab bhi koi incident hota hai, hum evidence file banate hain, matlab us waqt ki image aur video clip. Enquiry mein ya court mein sabse pehla sawaal yahi hota hai ki yeh footage beech mein badli to nahi gayi.

To agla step yeh hai: har evidence file ka SHA-256 hash banao, aur un hashes ki ek chain banao jahan har record ke andar pichle record ka hash ho. Isse agar koi beech ka ek frame ya ek record chhed deta hai to poori chain toot jaati hai aur turant pata chal jaata hai. Yeh hamare roadmap mein hai, abhi code mein nahi hai.

Yeh mat karna

- Yeh mat bolna ki blockchain implement kiya hua hai. Agar mentor ne code maang liya to position bahut kharab ho jayegi.
- Yeh bhi mat bolna ki blockchain zaroori nahi hai. Isse lagta hai ki aap theme se bhaag rahe ho.
- Sahi tareeka yeh hai: jo bana hai woh confidence se batao, jo nahi bana uska plan batao.

TAYYARI

Mentor Ke Sawaal Aur Unke Jawaab

Yeh sawaal mentor round mein sabse zyada aate hain. Jawaab apne shabdon mein bolna, lekin jo number aur wajah likhi hai woh wahi rakhna.

Q1. Isme naya kya hai? YOLO to aaj kal sab use karte hain.

Sir, model naya nahi hai, model ka istemaal naya hai. Main ek asli dikkat batata hoon jo humne solve ki.

Har camera alag height par aur alag angle par laga hota hai. Isliye ek hi aadmi ek camera mein bada dikhta hai aur doosre mein chhota. Ab agar hum speed ya doori pixel mein naapein to har camera ke liye alag calibration karni padegi. Do hazaar camera ke liye yeh mumkin hi nahi hai.

Humne isko aise solve kiya: hum har cheez us aadmi ki apni body height ke hisaab se naapte hain, pixel mein nahi. Chalna matlab lagbhag 0.45 body height per second, daudna matlab lagbhag 1.9. Yeh number har camera par same rehta hai, chahe camera teen meter par laga ho ya aath meter par.

Iska nateeja yeh hai ki naye camera par koi setup, koi calibration, koi measurement nahi chahiye. Stream jodo aur system chal padta hai. Yeh hi hamara asli innovation hai.

Q2. Naya camera lagane par model dobara train karna padega?

Nahi sir. Do wajah se.

Ek, detection model general hai, woh aadmi aur gaadi pehchanta hai, kisi ek camera ke liye train nahi hua.

Do, jaise maine bataya, hamara measurement body height par based hai, camera ke pixel par nahi. Isliye naya camera plug karne par bhi wahi rules lagte hain.

Q3. Raat mein kaam karega?

Sir, raat sabse badi dikkat hai aur main iska seedha jawaab deta hoon.

Abhi hamare paas low light detection ka module hai. Woh pehchaan leta hai ki frame kitna andhera hai aur uske hisaab se image ko enhance karke detection karta hai.

Lekin poori imaandaari se, bilkul andhere mein normal camera ki apni limit hai. Iska asli hal thermal camera hai. Hamari strategy RGB aur thermal ko milane ki hai, aur hamara architecture modular hai to yeh module jud sakta hai. Par abhi woh bana nahi hai kyunki test ke liye thermal camera nahi tha.

Q4. False alarm kitne aayenge? Jawaan pareshaan ho gaya to system band kar dega.

Sir, yeh sawaal hamare liye sabse important hai, aur hum bhi yahi maante hain ki zyada false alarm system ko maar deta hai.

Isliye humne ek usool rakha hai: ek frame par kabhi faisla nahi hota. Koi bhi track tab tak ginti mein nahi aata jab tak woh kam se kam teen baar lagataar dikhe.

Boundary ke paas ki harkat mein bhi do step hain. Pehle system sirf yeh note karta hai ki koi fence ke paas aa raha hai, us par alert nahi jaati. Alert tab jaati hai jab agle frames mein woh confirm hota hai.

Aur teesra, threat score ke chaar level hain. Har cheez alert nahi banti. LOW wali cheezein sirf log hoti hain, operator ko sirf upar wale level dikhte hain.

Q5. Border par internet nahi hota, cloud kaise chalega?

Sir, isiliye hamara design edge par chal sakta hai. Poora AI BOP ki apni machine par chal sakta hai aur alert wahin ban sakti hai. Internet ki zaroorat sirf tab hai jab headquarters ko summary bhejni ho, aur woh bhi sirf text aur chhoti image, poori video nahi.

Abhi hamara pura system ek hi machine par ek hi process mein chalta hai, jo edge deployment ka hi shape hai.

Q6. Ek machine par kitne camera chalenge?

Sir, abhi humne chaar camera ka setup banakar test kiya hai, ek hi machine par, bina GPU ke.

Speed ke liye humne do cheezein ki hain. Har doosra frame process hota hai, aur pose estimation sirf un logon par chalta hai jo frame mein kaafi bade dikh rahe hain.

GPU ke saath yeh number kaafi badh jaata hai, lekin main jhooth nahi bolunga, humne abhi bade scale ka load test nahi kiya hai.

Q7. Face recognition kar rahe ho? Privacy ka kya?

Sir, humne face recognition jaan bujhke scope se bahar rakha hai.

Hum sirf yeh dekhte hain ki koi insaan hai, woh kahan ja raha hai, aur kaise chal raha hai. Kaun hai yeh hum nahi dekhte aur na hi chehron ka koi database banate hain.

Iski wajah practical bhi hai. Border ke paas aam gaon wale bhi rehte hain aur roz aate jaate hain. Unke chehre ka database banana na zaroori hai na sahi.

Q8. Number plate kaise padhte ho, aur usse kya faayda?

Sir, gaadi detect hone par hum uske plate wale hisse ko crop karke OCR chalate hain.

Padhne ke baad hum do extra cheezein karte hain. Pehli, plate ke code se hum state aur district nikaalte hain. Hamare paas 41 state code aur lagbhag 998 district code ki table hai. To HR26 dikhe to system batata hai Gurgaon, Haryana. Border par yeh kaam ka hai, kyunki door ke state ki gaadi baar baar dikhna apne aap ek signal hai.

Doosri, hum plate ka rang dekhte hain. Safed matlab private gaadi, peela matlab commercial ya taxi, hara matlab electric. Isse gaadi ka type bina database ke pata chal jaata hai.

Imaandaari se ek limit bhi hai: har gaadi ka plate nahi padhta. Door ki, tedhi, ya dhundhli plate chhoot jaati hai. Jo padh jaata hai woh saaf record hota hai, aur jo nahi padhta uske liye system jhooth nahi bolta.

Q9. Test kaise kiya? Asli border footage hai kya?

Sir, asli border feed public mein nahi milti, isliye humne recorded video clips par test kiya hai jo border jaisi condition dikhate hain: fencing, khuli zameen, aur ek road checkpoint.

Software side par hamare paas 491 automated test hain jo har build par chalte hain. Yeh detection, tracking, behaviour, threat scoring, plate parsing aur login security, sabko cover karte hain.

Q10. Yeh market mein already nahi hai kya?

Sir, video analytics products market mein hain, lekin unmein do dikkat hain.

Ek, zyadatar unka apna camera ya apna box lena padta hai. Hamari poori shart yeh hai ki jo laga hai wohi chale.

Do, woh general purpose hote hain, jaise mall aur parking ke liye. Hamare rules border ke liye hain: fence ke paas rukna, ulti disha, raat ka time, group banana. Aur woh rules ek YAML file mein hain jo sector ke hisaab se badle ja sakte hain.

Q11. Aage kya plan hai?

Sir, teen cheezein line mein hain.

Pehli, evidence ke liye hash chain, jisse footage ke saath chhed chhaad pakdi ja sake.

Doosri, thermal camera ke saath low light ka poora hal.

Teesri, ek hi machine se badhakar kayi BOP ka central view, jahan har BOP par AI local chale aur summary upar jaye.

SIRF AAPKE LIYE

Kya Ban Chuka Hai Aur Kya Abhi Nahi Bana

Yeh page sirf aapke liye hai, mentor ko dikhane ke liye nahi. Isko padhkar jaana taaki aap galti se kuch aisa na bol do jo project mein hai hi nahi. Agar mentor ne kaha 'demo dikhao' aur cheez mili hi nahi, to usse zyada nuksaan kuch nahi karta.

Bana hua hai aur chal raha hai

- Object detection (YOLOv8n) aur tracking (ByteTrack) - bana hua hai aur chalta hai.
- Pose estimation (YOLOv8n-pose, 17 point) - bana hua hai.
- Body height ke hisaab se speed aur doori naapna - bana hua hai, aur yahi hamara sabse majboot innovation point hai.
- Boundary interaction ka arm-and-confirm logic - bana hua hai.
- Threat scoring, 16 factor, YAML se configure hone wala - bana hua hai.
- Anomaly detection (Isolation Forest) - bana hua hai.
- ANPR: plate padhna, state aur district nikalna, plate ke rang se gaadi ka type - bana hua hai.
- Evidence storage: incident ki image aur clip save hona - bana hua hai.
- Security: scrypt password hashing, session token ka fingerprint, do role, backend par permission check, aur 18 tarah ke audit log - bana hua hai.
- React dashboard jo FastAPI ke ek hi process se serve hota hai - bana hua hai.
- 491 automated test - chal rahe hain aur paas hain.

Abhi nahi bana

- Blockchain ya hash chain - nahi bana. Roadmap mein hai.
- RGB aur thermal fusion - nahi bana. Low light detection module hai, thermal nahi.
- Cloud ka hissa - abhi sab ek hi machine par chalta hai. Architecture modular hai lekin multi-BOP central deployment abhi nahi bana.
- Asli border ka live RTSP feed - abhi recorded video par test hua hai.
- Bade scale ka load test - chaar camera tak test hua hai, chalis par nahi.
- Face recognition - jaan bujhke nahi banaya, yeh scope se bahar hai.

EK NUMBER UPDATE

Ek chhota sa update: slide 3 par likha hai 460+ automated tests. Aaj ki tareekh mein yeh number 491 hai. Agar slide update kar sako to kar lo, warna bolte waqt 491 bol dena, kyunki zyada number aapke hi favour mein hai.

AAKHIR MEIN

Bolne Ke Tareeke Ki Kuch Baatein

Time ka hisaab	Slide 1 par 20 second, slide 2 par 2 minute, slide 3 par 2 se 3 minute, slide 4 par 2 minute, slide 5 par 1 se 2 minute, slide 6 par 30 second. Total lagbhag 9 minute, sawaal ke liye time bachega.
Shuruaat	Har slide par pehle ek line mein batao ki yeh slide kis baare mein hai, phir detail mein jao. Judge ko pata hona chahiye ki woh kya dekh raha hai.
Diagram	Slide 3 ka diagram baayen se daayen padho aur ungli se follow karo. Diagram par baat karte waqt point karna bahut farak daalta hai.
Jab jawaab na pata ho	Seedha bol do ki abhi yeh implement nahi kiya, aur phir batao ki kaise karte. Bahana banane se position kharab hoti hai, sach bolne se nahi.
Number yaad rakho	491 tests, 16 threat factors, 17 pose points, 41 state codes, 998 district codes, 2450 km border. Do teen asli number bol dene se lagta hai ki aapne khud banaya hai.
Team ka bantwara	Pehle se tay kar lo ki kaunsa slide kaun bolega, aur technical sawaal kaun lega. Beech mein ek doosre ko kaatna bura lagta hai.

> Aakhri baat. Judge ko perfect project se zyada woh team pasand aati hai jo apna project sach mein samajhti hai. Jo bana hai usko confidence se dikhao, jo nahi bana uska plan batao. Dono cheezein ek hi saans mein bol sakoge to round accha jayega.